

AI Marketing Campaign Management System

5-Part Backend Team Task Breakdown & Assignment Specification

This document divides the FastAPI & MySQL Backend Development into **5 independent, parallel modules**. Each team member is assigned full ownership of their specific module including Database Models, Pydantic Schemas, REST API Endpoints, and Initial Seed Data.

■ TEAM MEMBER 1: Core FastAPI Infrastructure & Profile Module

Module Scope: Core Server Setup, DB Engine Config & User Profile API

Target Files	backend/main.py, backend/database.py, backend/models.py
Database Table	profiles table (id, name, email, company, phone, industry, avatar)
API Endpoints	GET /api/profile, PUT /api/profile
Key Deliverables	• Initialize FastAPI app, Uvicorn runner, CORS middleware. • Configure SQLAlchemy MySQL connection string & SQLite fallback. • Build Profile model & Pydantic schemas with _id alias. • Seed default Sarah Jenkins profile on database startup.

■ TEAM MEMBER 2: Marketing Campaign Engine (CRUD)

Module Scope: Campaign Management System & Analytics Data

Target Files	backend/models.py (Campaign), backend/schemas.py, backend/main.py
Database Table	campaigns table (id, name, objective, status, budget, roi, startDate, endDate, channels)
API Endpoints	GET /api/campaigns, POST /api/campaigns, PUT /api/campaigns/{id}, DELETE /api/campaigns/{id}
Key Deliverables	• Implement Campaign model & JSON column support for channels. • Build full Campaign CRUD API endpoints. • Ensure Pydantic serializes _id string for frontend compatibility. • Seed 4 default marketing campaigns on database initialization.

■ TEAM MEMBER 3: Social Media Posts & Content Engine

Module Scope: Social Media Content Post Management

Target Files	backend/models.py (Post), backend/schemas.py, backend/main.py
Database Table	posts table (id, title, platform, content, imageUrl, publishedAt)
API Endpoints	GET /api/posts, POST /api/posts, DELETE /api/posts/{id}
Key Deliverables	• Build Post model with TEXT content & image URL storage. • Implement GET /api/posts sorted by publishedAt desc. • Implement post creation and deletion routes with validation.

■ TEAM MEMBER 4: Notifications & Audit History Logs Engine

Module Scope: Alerts, System Notifications & Audit Logging

Target Files	backend/models.py (Notification & History), backend/schemas.py, backend/main.py
Database Tables	notifications table & history_logs table
API Endpoints	GET /api/notifications, POST /api/notifications, PUT /api/notifications/mark-read, PUT /api/notifications/{id}/read GET /api/history, POST /api/history
Key Deliverables	• Implement Notification model (read/unread status, type formatting). • Implement History model for activity tracking. • Build mark-all-read & single notification read update endpoints. • Seed initial notifications & history logs.

■ TEAM MEMBER 5: n8n AI Content Proxy & API Integration Testing

Module Scope: AI Webhook Automation & Swagger Testing

Target Files	backend/main.py (n8n proxy), backend/schemas.py, backend/.env
Integration Target	n8n Automation Cloud Webhooks (Single & Batch content generation)
API Endpoints	POST /api/generate-ai-post, POST /api/generate-ai-batch
Key Deliverables	• Setup async HTTP client (httpx) for zero-CORS proxying. • Implement single post AI generation endpoint. • Implement batch multi-platform content generation endpoint. • Perform full OpenAPI / Swagger UI (http://localhost:5000/docs) integration testing.

Task Assignment Summary Matrix

Member	Module Name	Main Responsibility	APIs Handled
Member 1	Core Infrastructure & Profile	FastAPI, DB Engine, Profile API	2 Endpoints
Member 2	Campaign Engine	Campaign CRUD & Analytics	4 Endpoints
Member 3	Social Posts Engine	Post Creation & Media Assets	3 Endpoints
Member 4	Notifications & Audit Log	Alerts, Read Status, Activity Logs	6 Endpoints
Member 5	n8n AI Proxy & Testing	AI Generation & Swagger Testing	2 Endpoints